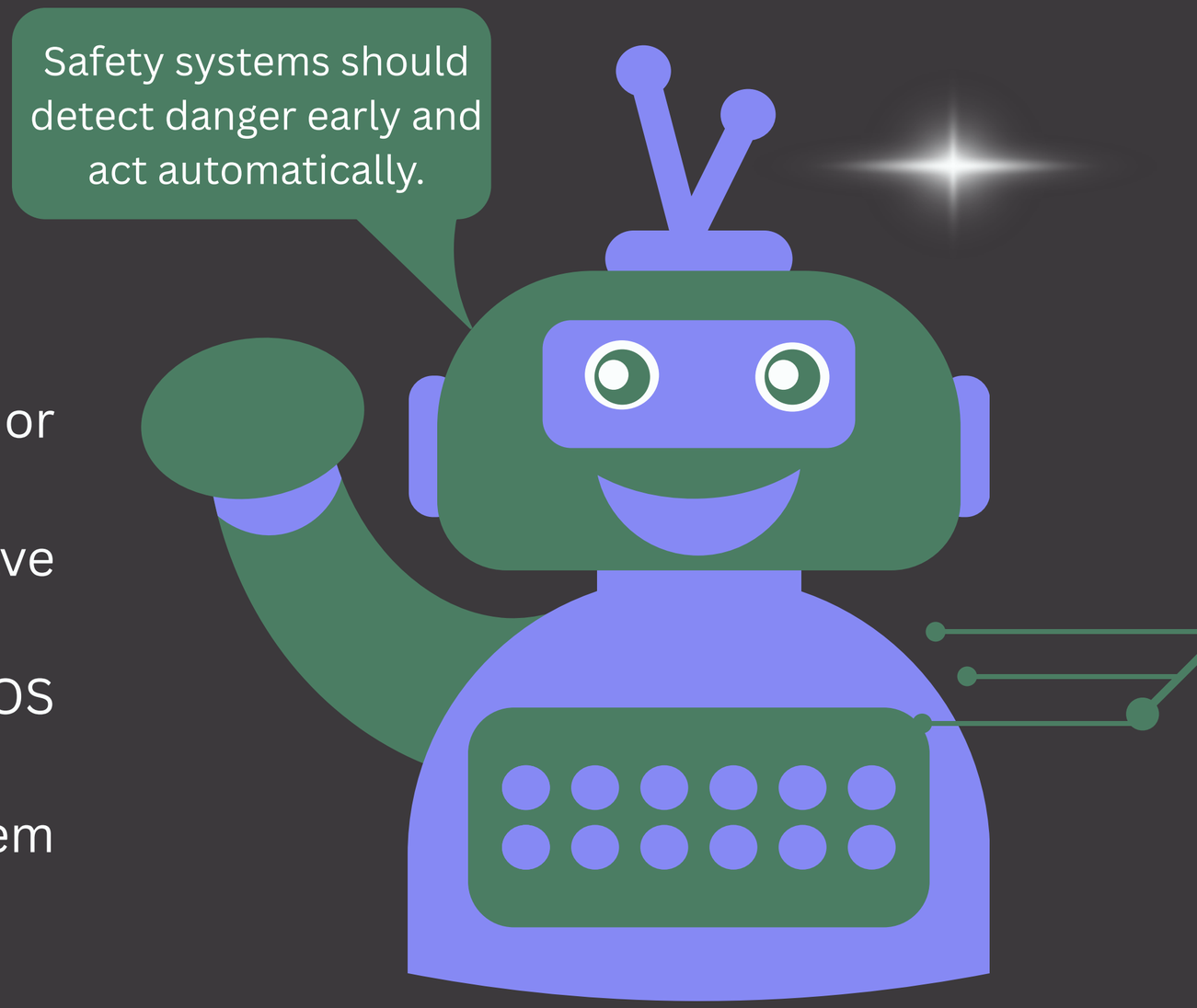
SAFETY COMPANION

-AI Powered Personal Safety Assistant

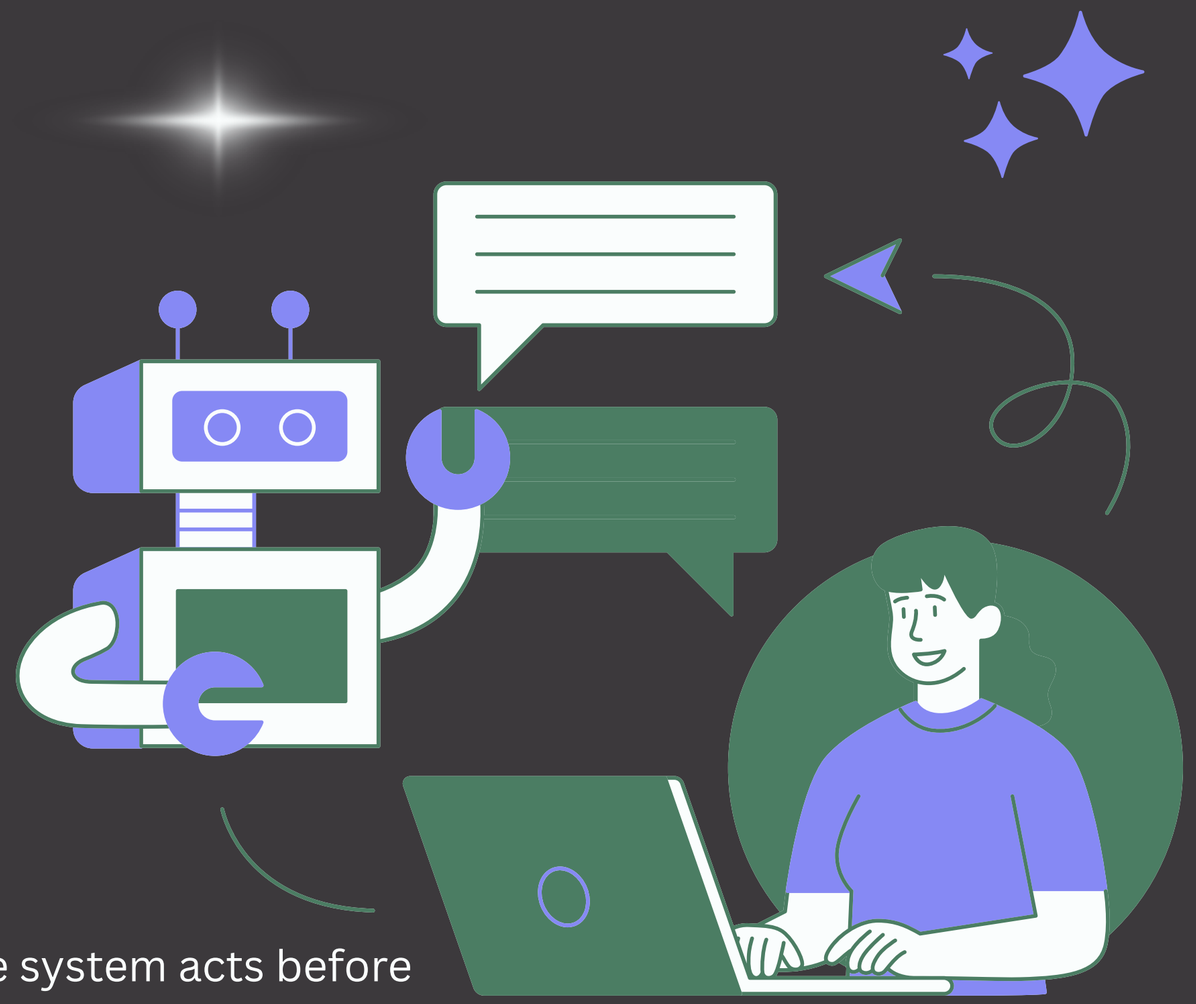
Problem statement

WHY?

- Many people walk alone in unfamiliar or unsafe environments.
- In dangerous situations, users may not have time to manually call for help.
- Current safety apps rely on manual SOS buttons, which may be too late.
- There is a need for a proactive safety system that can detect risk automatically.



SOLUTION



The system acts before the user presses SOS.

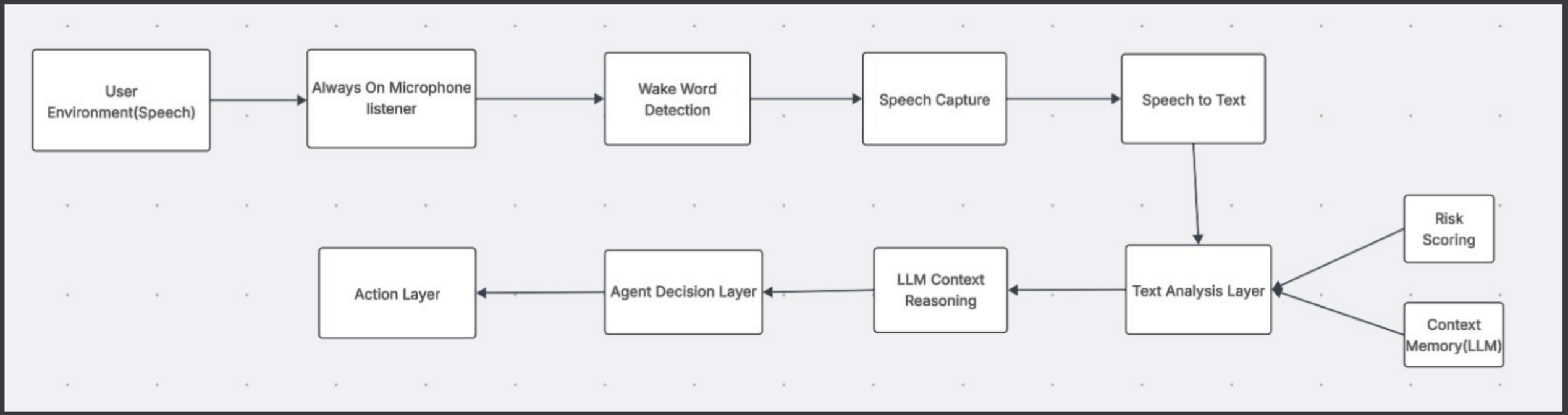
Safety Companion is an AI-powered background safety assistant that:

- Continuously listens for user voice input
- Analyzes stress, emotion, and context
- Detects suspicious or dangerous situations
- Uses an agentic decision system to activate safety tools automatically

Example actions:

- Fake call
- Flashlight
- Location sharing
- Emergency alert
- Alarm trigger

Architecture



The system combines rule-based safety detection with an LLM agent to reason about context and choose appropriate safety actions.

KEY FEATURES & IMPACT

Key Features

- Real-time voice safety monitoring
- Hybrid AI system (rule-based + LLM reasoning)
- Context-aware agentic decision making
- Automatic safety tool activation
- Continuous location sharing during danger
- Works as a background safety companion

Impact

- Helps users stay safe in uncertain environments
- Detects danger earlier than manual SOS apps
- Provides automatic emergency response

FUTURE IMPROVEMENTS

Future improvements

- Mobile app integration
- Smartwatch support
- Better environmental sensing
- Improved context reasoning

THANK YOU